#FLG1 GND PWR_FLAG
#FLG2 +24V PWR_FLAG
#FLG3 VSYS PWR_FLAG
#FLG4 DOOR_5V PWR_FLAG

Power

A diode guards against reversed 24 V. The buck feeds the matrix and the logic.

J1 1 IN_24V, 2 GND — W08: 24 V in, fuse F2

D1 1N5822: IN_24V (2) → +24V (1)

C1 100uF 35 V: +24V (1) — GND (2)

U1 D36V50F5, 5 V
- +24V (1), +24V (2) → VIN
- EN (3), GND (4), GND (5)
- VOUT → +5V (6), +5V (7)
- PG (8)

C2 470uF 16 V: +5V (1) — GND (2)

Pico 2

The diode lets USB and the board's 5 V both feed it.

U2 Raspberry Pi Pico 2

1	UART_TX	VBUS	40
2	SLIDE_UART	VSYS	39 VSYS
3	GND	GND	38 GND
4	SLIDE_STEP	3V3_EN	37
5	SLIDE_DIR	3V3	36 +3V3
6	SLIDE_EN	ADC_VREF	35
7	SLIDE_HOME	GP28	34 SPARE_GP28
8	GND	AGND	33 GND
9	DOOR_SENSE	GP27	32 SPARE_GP27
10	SPARE_GP7	GP26	31 SPARE_GP26
11	I2C_SDA	RUN	30 RUN
12	I2C_SCL	GP22	29 SPARE_GP22
13	GND	GND	28 GND
14	MATRIX_SCK	GP21	27 PORT_AT
15	MATRIX_TX	GP20	26 PORT_SET_3V3
16	LED1_EN	GP19	25 EXPO2
17	LED2_EN	GP18	24 EXPO1
18	GND	GND	23 GND
19	LED3_EN	GP17	22 TRIG2
20	LED4_EN	GP16	21 TRIG1

D2 1N5817: +5V (2) → VSYS (1)

J15 Spare pins
1 +3V3, 2 GND, 3 SPARE_GP7, 4 SPARE_GP22, 5 SPARE_GP26, 6 SPARE_GP27, 7 SPARE_GP28, 8 RUN

Filter slider

UART address 0. EN's pull-up keeps it off while the Pico starts.

U9 BTT TMC2209 V1.3

9	SLIDE_EN	VS	8 +24V
10	GND	GND	7 GND
11	GND	A2	6 MOTOR_A2
12	SLIDE_UART	A1	5 MOTOR_A1
13		B1	4 MOTOR_B1
14		B2	3 MOTOR_B2
15	SLIDE_STEP	VIO	2 +3V3
16	SLIDE_DIR	GND	1 GND

C5 100uF 35 V: +24V (1) — GND (2)

R19 10k: SLIDE_EN — +3V3

R20 1k: UART_TX — SLIDE_UART

J10 W28: slider's motor
1 MOTOR_A1, 2 MOTOR_A2, 3 MOTOR_B1, 4 MOTOR_B2

J11 W29: slider's home switch
1 HOME_SW, 2 GND

R21 10k: HOME_SW — +3V3

R22 1k: HOME_SW — SLIDE_HOME

C6 100nF: SLIDE_HOME — GND

Interlock and LED drivers

The door's switch alone powers U3: door open, every DIM falls to its pull-down.

J2 W35: door switch
1 DOOR_FEED, 2 DOOR_5V

F1 PTC, 0.1 A hold: +5V — DOOR_FEED

C3 100nF: DOOR_5V — GND

R1 3.9k: DOOR_5V — DOOR_SENSE

R2 6.8k: DOOR_SENSE — GND

U3 74AHCT125

1	GND	VCC	14 DOOR_5V
2	LED1_EN	1Y	3 DIM1
4	GND	2Y	6 DIM2
5	LED2_EN	3Y	8 DIM3
9	LED3_EN	4Y	11 DIM4
13	GND		
12	LED4_EN		
7	GND		

R3 4.7k: LED1_EN — GND

R4 1k: DIM1 — GND

U4 LDD-1000H
- +24V (23), +24V (24) → +VIN
- GND (1), GND (2) → −VIN
- DIM (22)
- +VOUT → LED1+ (13), LED1+ (14)
- −VOUT → LED1− (11), LED1− (12)

J3 W23: 395 nm
1 LED1+, 2 LED1−

R5 4.7k: LED2_EN — GND

R6 1k: DIM2 — GND

U5 LDD-1000H
- +24V (23), +24V (24) → +VIN
- GND (1), GND (2) → −VIN
- DIM (22)
- +VOUT → LED2+ (13), LED2+ (14)
- −VOUT → LED2− (11), LED2− (12)

J4 W24: 470 nm
1 LED2+, 2 LED2−

R7 4.7k: LED3_EN — GND

R8 1k: DIM3 — GND

U6 LDD-1000H
- +24V (23), +24V (24) → +VIN
- GND (1), GND (2) → −VIN
- DIM (22)
- +VOUT → LED3+ (13), LED3+ (14)
- −VOUT → LED3− (11), LED3− (12)

J5 W25: 565 nm
1 LED3+, 2 LED3−

R9 4.7k: LED4_EN — GND

R10 1k: DIM4 — GND

U7 LDD-1000H
- +24V (23), +24V (24) → +VIN
- GND (1), GND (2) → −VIN
- DIM (22)
- +VOUT → LED4+ (13), LED4+ (14)
- −VOUT → LED4− (11), LED4− (12)

J6 W26: 625 nm
1 LED4+, 2 LED4−

Cameras

Open-collector triggers, and exposure inputs with pull-ups.

R23 4.7k: TRIG1 — GND

R24 2.2k: TRIG1 — TRIG1_BASE

Q1 2N3904: TRIG1_BASE — TRIG1_OUT; GND

J12 W31: mono camera
1 TRIG1_OUT, 2 GND, 3 EXPO1_IN, 4 GND

R25 4.7k: EXPO1_IN — +3V3

R26 1k: EXPO1_IN — EXPO1

R27 4.7k: TRIG2 — GND

R28 2.2k: TRIG2 — TRIG2_BASE

Q2 2N3904: TRIG2_BASE — TRIG2_OUT; GND

J13 W32: colour camera
1 TRIG2_OUT, 2 GND, 3 EXPO2_IN, 4 GND

R29 4.7k: EXPO2_IN — +3V3

R30 1k: EXPO2_IN — EXPO2

LED matrix and camera port

5 V signals, through a buffer that's always on.

U8 74AHCT125

1	GND	VCC	14 +5V
2	MATRIX_SCK	1Y	3 MATRIX_SCK_5V
4	GND	2Y	6 MATRIX_TX_5V
5	MATRIX_TX	3Y	8 PORT_SET_5V
10	GND	4Y	11
9	PORT_SET_3V3		
13	+5V		
12	GND		
7	GND		

C4 100nF: +5V — GND

R11 4.7k: MATRIX_SCK — GND

R12 47: MATRIX_SCK_5V — MATRIX_CLK

R13 4.7k: MATRIX_TX — GND

R14 47: MATRIX_TX_5V — MATRIX_DATA

R15 4.7k: PORT_SET_3V3 — GND

R16 47: PORT_SET_5V — PORT_SET

J7 W30: LED matrix
1 +5V, 2 GND, 3 MATRIX_CLK, 4 MATRIX_DATA

J8 W33: port, DIG I/O 1
1 PORT_SET, 2 GND

J9 W34: port, DIG I/O 2
1 PORT_AT_5V, 2 GND

R17 3.9k: PORT_AT_5V — PORT_AT

R18 6.8k: PORT_AT — GND

LED temperatures

Each thermistor divides 3.3 V against 10k, into the I2C ADC at 0x48.

U10 ADS1115

4	AIN1	VDD	8 +3V3
5	AIN2	SDA	9 I2C_SDA
6	AIN3	SCL	10 I2C_SCL
7	AIN4	ALERT	2
1	ADDR		
3	GND		

C7 100nF: +3V3 — GND

R31 4.7k: I2C_SDA — +3V3

R32 4.7k: I2C_SCL — +3V3

J14 W27: thermistors
1 TH1, 2 GND, 3 TH2, 4 GND, 5 TH3, 6 GND, 7 TH4, 8 GND

R33 10k 1%: +3V3 — TH1

R34 1k: TH1 — AIN1

C8 100nF: AIN1 — GND

R35 10k 1%: +3V3 — TH2

R36 1k: TH2 — AIN2

C9 100nF: AIN2 — GND

R37 10k 1%: +3V3 — TH3

R38 1k: TH3 — AIN3

C10 100nF: AIN3 — GND

R39 10k 1%: +3V3 — TH4

R40 1k: TH4 — AIN4

C11 100nF: AIN4 — GND

Drawn by electronics/carrier/make.py; change circuit.py, not this file

Sheet: /
File: carrier.kicad_sch

Title: OPTIMA carrier board

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 1/1